Type MS · Rectangle and shape problems with inequalities

Answer 1 of 6 · Question 2 part whole · 6 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

Question number		
2	$w(w+2)$ or $w+w+2+w+w+2$ oe	B1
	$w^2+2w>8$ or $4w+4<30$ oe	B1
	$w<\frac{13}{2}$ oe	B1
	$w^2+2w-8\{=0\} \Rightarrow (w-2)(w+4)\{=0\} \Rightarrow w=...$	M1
	cv: $w=2$	A1
	$2<w<\frac{13}{2}$ oe	B1ft
		[6]
	ALT Works in terms of the length where $l=w+2$ or $w=l-2$	
	$l(l-2)$ or $l+l-2+l+l-2$ oe	B1
	$l^2-2l>8$ or $4l-4<30$ oe	B1
	$\left(l<\frac{34}{4}\right) \Rightarrow w<\frac{13}{2}$ oe	B1
	$l^2-2l-8\{=0\} \Rightarrow (l-4)(l+2)\{=0\} \Rightarrow l=...$	M1
	$l=4$	A1
	$\left(4<l<\frac{17}{2}\right) \Rightarrow 2<w<\frac{13}{2}$ o.e. in terms of w only.	B1ft
		[6]
		Total 6 marks

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Answer 1 of 6 · Question 2 part whole · 6 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

Mark	Additional Guidance **Allow working in other variables, such as x for all marks except the final A mark**
B1	For either correct expression shown.
B1	For either correct inequality shown (can be implied by a fully correct answer).
B1	For $w < \frac{13}{2}$, allow $=$ or any inequality
M1	Correct method to solve their 3TQ and finds at least one critical value of w (see general guidance for minimally acceptable attempt to solve).
A1 (M1 on ePen)	The correct positive cv solution ($w = 2$ or $w > 2$ or $w < 2$) from a correct 3TQ inequality or 3TQ equation, could be implied by their correct final answer, ignore the negative critical value even if it is incorrect.
B1ft (A1 on ePen)	Correctly combine their solutions from their linear and quadratic inequalities. Follows through their $\frac{13}{2}$ with their 2, both w values must be positive. Must be in terms of w.
ALT works in terms of l	
B1	For either correct expression shown
B1	For either correct inequality shown (can be implied by a fully correct answer)
B1	For $\frac{13}{2}$ in terms of w only UNLESS the final inequality is given as $2 < w < \frac{13}{2}$ in which case score B1 here.
M1	Correct method to solve their 3TQ equation or inequality and finds at least one critical value of w (see general guidance for minimally acceptable attempt to solve).
A1 (M1 on ePen)	For a value of $l = 4$ coming from a correct 3TQ Ignore the negative value even if it is incorrect.
B1ft (A1 on ePen)	Correctly combine their solutions from their linear and quadratic inequalities. Follows through their $\frac{13}{2}$ with their 2, both w values must be positive. Must be in terms of w

Type MS · Rectangle and shape problems with inequalities

Answer 1 of 6 · Question 2 part whole · 6 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Rectangle and shape problems with inequalities

Answer 2 of 6 · Question 4 part a · 2 marks

2022 JUN · Paper 01 (regional R) · find part a in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Rectangle and shape problems with inequalities

Answer 3 of 6 · Question 4 part b · 5 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Number		
4 (a) (i)	$4x + 8$ oe e.g. accept $x+(x+4)+x+(x+4)$	B1
(ii)	$x\,(x + 4)$ *or* $x^2 + 4x$	B1 [2]
(b)	$4x + 8 < 30$ ***or*** $x^2 + 4x > 12$	M1
	$x < 5.5$ oe	B1
	$(x^2 + 4x - 12 > 0)$ $(x + 6)(x - 2)(> 0)$	M1
	$x = 2$ $[x = -6]$	A1
	$2 < x < 5.5$	B1ft [5]

Part	Mark	Additional Guidance
(b)	**Note: Do not** accept $\leqslant$ or $\geqslant$ in place of < and >	
	M1	**Either** their linear expression < 30 (seen or implied in later work) ***or*** their quadratic expression > 12 (seen or implied in later work).
	B1	$x < 5.5$ oe allow implied M1 if $(4x + 8)$ not seen in earlier work. Allow $x = 5.5$ for this mark.
	M1	A minimally acceptable (see general guidance) attempt to factorise, use the formula or complete the square to solve their 3TQ.
	A1	For the correct critical value $x = 2$. Ignore $x = -6$ for this mark. This may be given as an inequality. Award the mark for 2 seen even if it is embedded in a spurious inequality.
	B1ft	ft their values for x, provided the inequality is written correctly and is correct for their values. Must be in the form $a < x < b$ Both values of x must be positive. Their value of '2' must be less than their value from the linear solution. i.e < their 5.5

Type MS · Rectangle and shape problems with inequalities

Answer 3 of 6 · Question 4 part b · 5 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Rectangle and shape problems with inequalities

Answer 4 of 6 · Question 5 part whole · 7 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

Number		
5	$(P=)\ 6x+6$ $(A=)\ x(2x+3)$ $6x+6>10$ or $x(2x+3)<35$ $x>\frac{2}{3}$ oe $\left(2x^2+3x-35<0\right)$ $(2x-7)(x+5)(<0)$ (critical values are) $x=\frac{7}{2}$, $(x=-5)$ $\frac{2}{3}<x<\frac{7}{2}$	B1 B1 M1 A1 M1 A1 A1ft [7]
		Total 7 marks

Type MS · Rectangle and shape problems with inequalities

Answer 4 of 6 · Question 5 part whole · 7 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Rectangle and shape problems with inequalities

Answer 5 of 6 · Question 2 part a · 1 mark

2022 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

	$\frac{}{2}$ is sufficient working
A1	For $19-11\sqrt{3}$ (Allow $a=19,\ b=-11$) As this result can be achieved with a calculator, there can be no incorrect working shown for this mark to be awarded.
ALT	
M1	For $\left(3\sqrt{3}+5\right)\left(a+b\sqrt{3}\right)$
dM1	For attempting to multiply out $\left(3\sqrt{3}+5\right)\left(a+b\sqrt{3}\right)$ to reach an expression that can be used to compare coefficients. There may be up to 2 errors or omissions.
A1	For forming 2 correct simultaneous equations and solving correctly to reach $19-11\sqrt{3}$ (Allow $a=19,\ b=-11$). Two common examples are shown in the scheme. As this result can be achieved with a calculator, there can be no incorrect working shown for this mark to be awarded.

Type MS · Rectangle and shape problems with inequalities

Answer 5 of 6 · Question 2 part a · 1 mark

2022 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks
2 a	$(8x+y)(6x+y)\left[=48x^2+14xy+y^2\right]$ oe	B1 (1)
b	$48x^2+14xy+y^2-2(9x^2+15x^2)$ oe $=14xy+y^2$ *	M1 M1 A1 cso* (3)
ALT	eg $3(y\times 3x)+(y\times 5x)+y^2$ or $y(8x+y)+2(3x\times y)$ or $y\times 3x+y\times 5x+y\times(6x+y)$ or $y(6x+y)+y(8x+y)-y^2$ $=14xy+y^2$ *	M1 M1 A1 cso* (3)
c	$(14xy+y^2=1080,\ 48x^2+14xy+y^2=3432)$ $48x^2+1080=3432$ $\left(48x^2=2352\right)$ $x=7$ $y^2+98y-1080(=0)$ $(y-10)(y+108)(=0)$ $y=10$	M1 A1 M1 M1 A1 (5)
ALT	$(14xy+y^2=1080,\ 48x^2+14xy+y^2=3432)$ $x=\dfrac{1080-y^2}{14y}\Rightarrow 48\left(\dfrac{1080-y^2}{14y}\right)^2+\left(\dfrac{1080-y^2}{14y}\right)14y+y^2=3432$ $\Rightarrow y^4-2160y^2+1166400=0$	M1
	$(y^2-1164)(y^2-100)(=0)$ $\left(\Rightarrow y^2=1164 \text{ or } y^2=100 \text{ or } y=\pm 108 \text{ or } y=\pm 10\right)$	M1 (A1 on ePen)
	$y = 10$	A1 (M1 on ePen)
	$14(10)x+100=1080\Rightarrow x=$	M1
	$x=7$	A1
		Total 9 marks

Type MS · Rectangle and shape problems with inequalities

Answer 6 of 6 · Question 2 part b · 3 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

	$\frac{}{2}$ is sufficient working
A1	For $19-11\sqrt{3}$ (Allow $a=19,\ b=-11$) As this result can be achieved with a calculator, there can be no incorrect working shown for this mark to be awarded.
ALT	
M1	For $\left(3\sqrt{3}+5\right)\left(a+b\sqrt{3}\right)$
dM1	For attempting to multiply out $\left(3\sqrt{3}+5\right)\left(a+b\sqrt{3}\right)$ to reach an expression that can be used to compare coefficients. There may be up to 2 errors or omissions.
A1	For forming 2 correct simultaneous equations and solving correctly to reach $19-11\sqrt{3}$ (Allow $a=19,\ b=-11$). Two common examples are shown in the scheme. As this result can be achieved with a calculator, there can be no incorrect working shown for this mark to be awarded.

Type MS · Rectangle and shape problems with inequalities

Answer 6 of 6 · Question 2 part b · 3 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks
2 a	$(8x+y)(6x+y)\left[=48x^2+14xy+y^2\right]$ oe	B1 (1)
b	$48x^2+14xy+y^2-2(9x^2+15x^2)$ oe $=14xy+y^2$ *	M1 M1 A1 cso* (3)
ALT	eg $3(y\times 3x)+(y\times 5x)+y^2$ or $y(8x+y)+2(3x\times y)$ or $y\times 3x+y\times 5x+y\times(6x+y)$ or $y(6x+y)+y(8x+y)-y^2$ $=14xy+y^2$ *	M1 M1 A1 cso* (3)
c	$(14xy+y^2=1080,\ 48x^2+14xy+y^2=3432)$ $48x^2+1080=3432$ $\left(48x^2=2352\right)$ $x=7$ $y^2+98y-1080(=0)$ $(y-10)(y+108)(=0)$ $y=10$	M1 A1 M1 M1 A1 (5)
ALT	$(14xy+y^2=1080,\ 48x^2+14xy+y^2=3432)$ $x=\dfrac{1080-y^2}{14y}\Rightarrow 48\left(\dfrac{1080-y^2}{14y}\right)^2+\left(\dfrac{1080-y^2}{14y}\right)14y+y^2=3432$ $\Rightarrow y^4-2160y^2+1166400=0$	M1
	$(y^2-1164)(y^2-100)(=0)$ $\left(\Rightarrow y^2=1164 \text{ or } y^2=100 \text{ or } y=\pm 108 \text{ or } y=\pm 10\right)$	M1 (A1 on ePen)
	$y=10$	A1 (M1 on ePen)
	$14(10)x+100=1080\Rightarrow x=$	M1
	$x=7$	A1
		Total 9 marks